Results Overview

Excel

The descriptive analysis provides an overview of housing burden and household insurance expenditures across 91,991 observations. Through summary statistics, pivot tables, and charts, the analysis examines the distribution of housing burden, differences in expenditure levels, and variations across demographic and housing characteristics. The results show variation in both housing burden and expenditure patterns, highlighting several areas for further research.

The summary statistics indicate that the average housing burden is approximately 20.27%, compared to the median of 15%. The gap between the mean and median suggests that housing burden is not evenly distributed across the sample. Although most observations have a housing burden below the overall average, some have much higher burdens, raising the average. Housing burden ranges from 0% to 100%, demonstrating variation in the proportion of income allocated to housing. The distribution of housing burden is important because households facing different levels of housing costs relative to their income have different financial circumstances. Examining housing burden across household characteristics provides additional context for understanding these differences. However, the descriptive statistics alone do not establish why some households experience higher housing burdens than others.

The pivot tables show that average housing burden varies across housing tenure categories. The largest tenure group, category 1 (owned with mortgage), contains 36,072 observations and has an average housing burden of approximately 19.06%. Category 2 (Owned without mortgage) contains 24,487 observations and has a lower average of approximately 16.29%. In contrast, category 4 (Rented) contains 30,150 observations and has an average housing burden of approximately 25.29%, the highest among the five reported categories. Categories 5 (Occupied w/o payment of cash rent) and 6 (Student Housing) have lower average housing burdens of approximately 13.38% and 4.88%. However, these categories contain substantially fewer observations, particularly category 6. Consequently, their averages should be interpreted cautiously and should not be assumed to represent patterns that apply equally to the broader population. The differences across tenure categories demonstrate that housing burden varies considerably across the housing circumstances represented in the dataset. The pivot tables also show variation across marital status, race, sex, education, and state categories.

The summary statistics reveal substantial differences in average expenditure levels across the four insurance categories represented in the expenditure pivot tables. Average adjusted personal insurance expenditure is approximately $1,511.74, making it the largest of the four categories examined, despite life insurance being included in that calculation. Health insurance has an average adjusted expenditure of approximately $652.69, followed by vehicle insurance at $270.27 and life insurance at $93.28. These differences indicate that the recorded expenditure

amounts vary considerably by category. Personal insurance has a substantially higher average expenditure than life insurance, while health insurance falls between personal and vehicle insurance. The median values provide additional context for interpreting these averages. For example, the median adjusted personal insurance expenditure is $766, compared with a mean of approximately $1,512. Similarly, adjusted health insurance expenditure has a median of $415 and a mean of approximately $653. The differences between the means and medians indicate that expenditure distributions are not symmetrical and that relatively large expenditure values influence the averages. The number of zero-expenditure observations also differs substantially across categories. Approximately 49.0% of observations have zero vehicle insurance expenditure, 20.6% have zero health insurance expenditure, 17.5% have zero personal insurance expenditure, and 75.1% have zero life insurance expenditure. These patterns are important when interpreting the averages because the reported means combine observations with zero expenditure and observations with positive expenditure. In particular, the high proportion of zero life insurance expenditures suggests that the overall average does not describe the spending behavior of every household equally well. Differences in average expenditures may reflect differences in both the likelihood of recording a positive expenditure and the amounts recorded among observations with positive expenditures.

The expenditure pivot tables further demonstrate that average insurance expenditures differ across housing tenure categories. For category 1 (owning with mortgage), average adjusted vehicle insurance expenditure is approximately $330.50, health insurance expenditure is $774.76, personal insurance expenditure is $2,256.07, and life insurance expenditure is $135.17. In category 2 (owning without mortgage), the corresponding averages are approximately $243.36, $806.79, $1,089.16, and $94.50. Category 4 (Rented) has an average adjusted vehicle insurance expenditure of approximately $223.91, health insurance expenditure of $395.04, personal insurance expenditure of $997.61, and life insurance expenditure of $43.84. Compared with category 1, category 4 has lower average expenditure in all four categories.

The differences are particularly pronounced for personal insurance. The average adjusted personal insurance expenditure in category 1 (owning with mortgage) is more than twice that in category 4 (Rented). Health insurance expenditures also differ substantially, with category 2 (owning without mortgage) having a higher average than category 1 (owning with mortgage), while category 4 (Rented) has a considerably lower average than either group. These comparisons show that expenditure patterns vary across housing tenure categories and that the differences are not identical for every insurance outcome. However, the tenure codes must be interpreted using the dataset's original documentation before assigning labels such as owner, renter, or mortgage holder.

Taken together, the summary statistics, pivot tables, and charts provide evidence of substantial variation in housing burden and insurance expenditures across the dataset. Three findings stand out. First, the difference between mean and median housing burden indicates that the distribution

includes observations with considerably higher burdens than the median. Second, personal insurance has the highest average expenditure among the four insurance categories examined, whereas life insurance has the lowest average and the largest proportion of zero-expenditure observations. Third, average expenditures differ across housing tenure categories, with particularly large differences in personal and health insurance expenditures. These descriptive findings provide a foundation for the next stage of the project, which investigates the relationship between housing burden and insurance expenditures using statistical models. The descriptive analysis identifies differences and patterns worth examining further.

SQL

The same dataset was used in the Excel analysis and the SQL analysis. The main goal of the SQL analysis was to confirm what was found in Excel and to provide additional information. Like the dataset used in Excel, there are 91,991 observations and 30+ variables from the Consumer Expenditure Survey. The analysis focused on the same variables: housing burden, insurance expenditures, household characteristics, and differences across housing tenure groups. Overall, the results indicate that housing burden and insurance expenditures vary considerably across households.

The same results were found of an average of approximately 20.3%, and median of 15%, which suggests this data is positively skewed, which means some higher housing burden values raise the overall average. Housing burden ranged from 0 to 1. 0.72% (663 observations) of the sample have a housing burden of zero. Overall, the results demonstrate substantial variation in housing burden across the sample and suggest that the median provides useful context alongside the mean. The data about the average insurance expenditures were confirmed. Personal insurance had the highest average expenditure among the four insurance categories examined, at approximately $1,512, followed by health insurance at $653, vehicle insurance at $270, and life insurance at $93. The means were higher than the medians for vehicle, health, and personal insurance, indicating that higher expenditure observations influence the averages, which Excel also showed. Life insurance had a median of zero because approximately 75% of households reported no expenditure in that category. Vehicle insurance also had a substantial proportion of zero values, with approximately 49% of observations reporting no expenditure. These numbers line up with Excel.

The housing tenure results also line up with Excel. Among the three largest groups, the category coded as 4 had the highest average housing burden, at approximately 25.3%, followed by category 1 at 19.1% and category 2 at 16.3%. The difference between categories 2 and 4 was approximately nine percentage points. Categories 5 and 6 had lower average housing burdens, although these groups contained substantially fewer observations and should be interpreted carefully. The average insurance expenditures by housing tenure numbers also match up. Category 1 had the highest average vehicle insurance expenditure, at approximately $331,

compared with $243 for category 2 and $224 for category 4. Health insurance showed a different pattern, with category 2 recording the highest average expenditure at approximately $807, followed by category 1 at $775 and category 4 at $395. Personal insurance expenditure was substantially higher in category 1, averaging approximately $2,256, compared with $1,089 in category 2 and $998 in category 4. Life insurance expenditure was also highest in category 1. These results indicate that no single tenure category consistently had the highest average expenditure across all insurance types.

There was some additional analysis with other expenditure variables. Food expenditure was positive for nearly all observations, while alcoholic beverage expenditure, vehicle financing, and mortgage expenditure contained substantial proportions of zero values. For example, approximately 51% of observations reported zero alcoholic beverage expenditure, 68% reported zero vehicle financing expenditure, and 57% reported zero mortgage expenditure. These values are not necessarily data errors, as households may legitimately have no spending in a category during the period measured. Housing burdens were also placed into groups. Between 0 and 0.3 was classified as lower burden, between 0.3 and 0.5 as moderate burden, and anything above 0.5 as Higher burden. Comparisons across housing-burden groups did not reveal a consistent increase or decrease in insurance expenditures as housing burden increased. Average vehicle insurance expenditure was highest among households with housing burden between 20% and 30%, while average health insurance expenditure was also highest in this group. Expenditures in these categories were lower among households with a housing burden above 50%. These patterns suggest that the relationship between housing burden and insurance expenditures may be more complex than a simple linear pattern. However, the grouped comparisons are descriptive and do not account for differences in other household characteristics, such as income, age, employment, or family size.

Several data characteristics should be considered when interpreting the results. First, the expenditure distributions contain many zero values, particularly for life insurance, vehicle financing, and mortgage expenditure. Second, the means of several expenditure variables are substantially higher than their medians, indicating uneven distributions. Third, some housing tenure categories contain far fewer observations than the largest groups, making comparisons involving those categories less reliable. Finally, the adjusted insurance variables add one to the original expenditure values, allowing logarithmic transformations when original expenditures are zero. Overall, the SQL analysis shows that housing burden and insurance expenditures vary considerably across households and that expenditure patterns differ across housing tenure groups. Personal insurance had the highest average expenditure among the insurance categories examined, while life insurance had a median of zero because most households reported no spending in that category. Housing burden also differed across tenure groups, with category 4 having a higher average than the other two largest groups. Comparisons across housing-burden groups did not demonstrate a consistent increase in insurance expenditure as housing burden

increased. These findings establish a descriptive baseline for the capstone project and identify patterns that can be examined further through statistical modeling.

Python

The same dataset is used so there are 91,991 observations and 30+ variables from the Consumer Expenditure Survey dataset. The purpose of the notebook was to validate the dataset, examine the distributions of housing burden and insurance expenditures, compare household characteristics across housing tenure categories, and estimate a regression model examining the relationship between housing burden and vehicle insurance expenditure. The initial data inspection confirmed that the dataset contained the expected variables, including housing burden, demographic characteristics, household expenditures, and adjusted insurance expenditure variables. This established a foundation for the subsequent descriptive and regression analyses.

The data validation process identified no missing values across the 34 variables in the dataset. This means that the variables included in the analysis had complete recorded values for all 91,991 observations. However, the absence of missing values does not necessarily mean that every observation is valid, as some variables may contain zero values or other codes that require interpretation. The duplicate-observation check identified two duplicated rows. Although this represents a very small proportion of the dataset, the duplicated records should be examined to determine whether they are legitimate repeated observations or unintended duplicates.

Housing burden averaged approximately 20.3%, with a median of 15%. The standard deviation was approximately 18.3 percentage points, indicating substantial variation in housing burden across households. The distribution ranged from 0% to 100%, while the middle 50% of observations fell between approximately 8% and 26%. The mean being higher than the median suggests that the distribution is positively skewed, with some households experiencing relatively high housing burdens. The notebook recorded 663 observations with a housing burden of zero, representing approximately 0.7% of the sample. These results indicate that housing burden varies considerably across households and that reporting both the mean and median provides a more complete description of the distribution.

Insurance expenditure variables exhibited substantial differences in their distributions. Average vehicle insurance expenditure was approximately $269, compared with a median of $101. Health insurance expenditure averaged approximately $652, with a median of $414. Personal insurance expenditure had the highest average among these categories, at approximately $1,511, with a median of $765. Life insurance expenditure averaged approximately $92, while its median was zero. The differences between the means and medians indicate that the expenditure distributions are uneven, with higher expenditure observations influencing the averages.

The analysis identified substantial numbers of zero values in several expenditure categories. Approximately 49% of households reported zero vehicle insurance expenditure, 20.6% reported zero health insurance expenditure, 17.5% reported zero personal insurance expenditure, and 75.1% reported zero life insurance expenditure. These values indicate that spending is not evenly distributed across households and that many households have no recorded expenditure in particular insurance categories. The dataset also contains adjusted insurance variables, including AdjVEHINSPQ, AdjHLTHINPQ, AdjPERINSPQ, and AdjLIFINSPQ. Each adjusted variable adds one to its corresponding original expenditure variable. This adjustment allows logarithmic transformations when the original expenditure is zero, although the resulting model describes the logarithm of expenditure plus one rather than the logarithm of expenditure alone.

The grouped descriptive analysis revealed differences in housing burden across the five observed CUTENURE categories. Category 1 contained 36,072 observations and had an average housing burden of approximately 19.1%. Category 2 contained 24,487 observations and had an average of 16.3%. Category 4 contained 30,150 observations and had the highest average among the three largest groups, at approximately 25.3%. Categories 5 and 6 had lower average housing burdens of approximately 13.4% and 4.9%, respectively, but contained substantially fewer observations. These results suggest that housing burden differs across tenure categories.

The notebook also compared descriptive statistics across the categories of QSQ118, which ranged from 0 to 23. Average housing burden was relatively similar across these groups, generally falling between approximately 19.6% and 20.8%. The median was usually between 14% and 15%, suggesting that the overall distribution of housing burden was broadly similar across the coded categories. Although some groups had slightly higher averages than others, the differences were modest compared with the differences observed across housing tenure categories.

The notebook estimated an ordinary least squares regression model in which the dependent variable was the natural logarithm of adjusted vehicle insurance expenditure, log(AdjVEHINSPQ). Housing burden was the primary explanatory variable, while the model controlled for age, housing tenure, family size, hours worked, marital status, race, sex, vehicle quantity, gasoline expenditure, vehicle financing, mortgage expenditure, rent, public transportation expenditure, state, vehicle ownership purpose, Hispanic status, education, and QSQ118. The model included 91,991 observations and had an R-squared of approximately 0.129 and an adjusted R-squared of 0.128. This means that the included explanatory variables accounted for approximately 12.9% of the observed variation in the logarithm of adjusted vehicle insurance expenditure. Although the model was statistically significant overall, its explanatory power was limited, indicating that other factors not included in the model may also be associated with differences in vehicle insurance expenditure.

Housing burden had a positive and statistically significant coefficient of approximately 2.113 in the vehicle insurance regression model, with a p-value below 0.001. Because the dependent variable is the logarithm of adjusted vehicle insurance expenditure, the coefficient indicates that higher housing burden was associated with higher adjusted vehicle insurance expenditure, holding the other included variables constant. Since housing burden is measured as a proportion from 0 to 1, a 0.01 increase in housing burden represents a one-percentage-point increase. Therefore, a 0.01 increase in housing burden is associated with approximately 2.11% higher adjusted vehicle insurance expenditure, holding the other included variables constant. This is a conditional association rather than evidence that housing burden causes vehicle insurance expenditure to increase. The result should also be interpreted in light of the model's assumptions and the possibility of omitted variables or heteroskedasticity.

The regression results also indicated differences in adjusted vehicle insurance expenditure across certain housing tenure categories. Relative to the reference category, category 4 had a negative and statistically significant coefficient of approximately -0.197, while categories 5 and 6 also had negative and statistically significant coefficients. Category 2 did not differ significantly from the reference category at conventional significance levels. Age had a small negative and statistically significant coefficient, while hours worked had a positive and statistically significant coefficient. Family size was not statistically significant in the estimated model. These findings indicate that some household characteristics were associated with adjusted vehicle insurance expenditure after accounting for the other included variables. The interpretation of the tenure coefficients depends on verifying the reference category and the official labels for the tenure codes.

The notebook calculated variance inflation factors to assess multicollinearity among the explanatory variables. Most continuous variables and many categorical indicators had relatively low VIF values. Housing burden had a VIF of approximately 1.19, suggesting that it was not strongly explained by a linear combination of the other model predictors. However, several education-category indicators had high VIF values, including values above 100 for some categories of HIGH_EDU. These results warrant further examination of the education variable's coding, category frequencies, and its relationship with other predictors. Because the model represents education as a categorical variable, the individual dummy-variable VIFs should be interpreted cautiously rather than treated as a direct measure of the entire education variable's multicollinearity. The high values do not automatically mean that the housing-burden coefficient is unreliable, but they indicate that the model's categorical predictors deserve additional review. Only the vehicle insurance model was checked, so the other models will have to be checked to see if similar results occur.

Overall, the Python validation analysis confirmed that the dataset contained 91,991 observations, had no recorded missing values, and contained two duplicated rows that should be reviewed. Descriptive statistics showed considerable variation in housing burden and insurance

expenditures, with substantial zero expenditure rates in several insurance categories. Housing burden also differed across housing tenure categories, while its average was comparatively similar across the QSQ118 groups. In the regression model, housing burden had a positive and statistically significant association with adjusted vehicle insurance expenditure after controlling for the other included variables. The model explained approximately 12.9% of the variation in the logged outcome, indicating that other factors likely contribute to differences in vehicle insurance expenditure. The multicollinearity assessment identified high VIF values for some education-category indicators, suggesting an area for further model evaluation. Taken together, these findings provide a useful validation and statistical analysis stage for the capstone, while highlighting the importance of checking duplicate records, verifying categorical codes, and evaluating regression assumptions before drawing stronger conclusions.

R

The R component of the capstone project focuses on regression analysis to examine the relationship between housing burden and four categories of adjusted insurance expenditures: vehicle, life, personal, and health insurance. The regression models use the natural logarithm of adjusted insurance expenditure as the dependent variable and include housing burden (`HOUSB`) as the primary explanatory variable. Additional predictors account for household characteristics, demographics, housing tenure, work hours, vehicle-related expenses, other expenditures, state, education, and survey-period categories. This analysis extends the descriptive work conducted in Excel, SQL, and Python by examining the association between housing burden and insurance expenditures while controlling for other included variables.

The original regression models produced different associations between housing burden and the four insurance expenditure outcomes. The coefficient on housing burden was positive in the vehicle insurance model (2.113) and life insurance model (0.482), but negative in the personal insurance model (-0.742) and health insurance model (-0.538). These results suggest that housing burden does not have a uniform association with all four insurance expenditure categories. The original vehicle insurance model had an R-squared of approximately 0.129, the life insurance model approximately 0.083, the personal insurance model approximately 0.542, and the health insurance model approximately 0.248. Among these models, the personal insurance model explained the largest proportion of variation in its dependent variable. However, these results describe conditional associations and do not establish that housing burden causes changes in insurance expenditure.

An additional step involved correcting the treatment of QSQ118 in the regression specifications. Treating this variable as a categorical factor using factor(QSQ118) allows the models to estimate separate coefficients for the represented categories rather than assuming that the numeric codes have a linear relationship with the outcome. The vehicle insurance model already treated QSQ118 as categorical, while the other models were updated to use this

specification. Following the correction, the personal insurance model's R-squared increased from approximately 0.542 to 0.544, and the health insurance model's R-squared increased from approximately 0.248 to 0.261. The life insurance model's R-squared remained approximately 0.083. These changes indicate that treating the survey-period variable as categorical affected the models to different degrees, particularly the health insurance model.

The regression diagnostics identified substantial evidence of heteroskedasticity, meaning that the variance of the model errors was not constant across observations. Breusch–Pagan tests returned very small p-values for the regression models

Power BI

The Power BI component of the capstone project presents housing burden and insurance expenditure data through an interactive dashboard. Its purpose is to make selected findings easier to explore and interpret while allowing users to compare observations across household characteristics and geographic categories. The dashboard contains four key performance indicator (KPI) cards, scatter plots examining relationships between housing burden and insurance expenditures, a comparison of selected vehicle-related expenditure measures, and slicers that allow users to filter the displayed results. This component emphasizes interactive data exploration rather than the statistical estimation performed in R.

The four KPI cards provide a concise overview of important measures in the dataset. They display average adjusted vehicle insurance expenditure, average housing burden, average family size, and the count of housing-burden observations. Together, these indicators establish a baseline for exploring the data. Average housing burden summarizes the proportion of income allocated to housing, while average adjusted vehicle insurance expenditure provides context for one of the project's primary insurance outcomes. Average family size adds information about household composition, and the observation count indicates how many records contribute to the displayed housing-burden measure. The count should be understood as a count of observations rather than unique households unless the underlying data confirms that each record represents a distinct household. Clear titles and appropriate formatting, including percentage formatting for housing burden and dollar formatting for expenditure, help users interpret the KPI values.

The scatter plots extend the dashboard beyond overall summary measures by examining housing burden in relation to adjusted vehicle insurance expenditure and adjusted health insurance expenditure. These visualizations allow users to explore how the two insurance outcomes vary across observations with different housing-burden levels. A scatter plot can reveal the overall pattern, the spread of observations, and the presence of unusually high or low values. However, expenditure data may be unevenly distributed, and overlapping observations can make some patterns difficult to see. The scatter plots are therefore useful for exploratory analysis, but they do not establish that housing burden causes insurance expenditure to increase or decrease.

They also complement the regression models by providing a visual perspective on relationships that R examines while controlling for additional variables.

Another dashboard visualization compares vehicle insurance expenditure with selected vehicle-related expenditure measures, including gasoline and motor oil and vehicle financing. This comparison provides context for examining different categories of vehicle-related spending. However, these measures represent distinct expenditure categories, and their averages may differ substantially in scale. They should not be treated as interchangeable measures of the same behavior. Clear titles, axis labels, and dollar formatting are important for making the comparison understandable. The visualization is best interpreted as a descriptive comparison of expenditure amounts rather than as evidence of a direct relationship between the categories.

The dashboard includes slicers for housing tenure (CUTENURE), state (STATE), family size (FAM_SIZE), marital status (MARITAL1), and QSQ118. These controls allow users to examine selected subsets of the data without creating separate static charts for every group. For example, users can filter the dashboard by housing-tenure category or state and examine how the displayed averages and scatter plots change. The slicers make Power BI particularly useful for exploring subgroup differences and investigating whether patterns visible in the overall dataset also appear within selected groups. Their interpretation depends on using appropriate category labels and ensuring that each slicer filters the intended visuals.

Power BI contributes a different type of analysis from the other tools used in the capstone. Excel provides summary statistics, pivot tables, and charts; SQL supports data querying and grouped calculations; Python supports data validation, descriptive analysis, and regression modeling; and R provides a more extensive examination of conditional relationships through regression. Power BI builds on these components by organizing selected measures and relationships into an interactive dashboard. Users can explore the data by changing filters and observing how the displayed results respond.

Overall, the Power BI dashboard provides an accessible way to examine housing burden, insurance expenditures, and household characteristics. Its main contribution is the ability to interact with the data and compare selected groups while maintaining a consistent visual layout. The KPI cards establish an overview, the scatter plots support exploration of relationships between housing burden and insurance expenditure, and the slicers make subgroup comparisons possible. Together, these features complement the statistical and descriptive work completed in the other sections of the capstone.

Tableau

The Tableau component of the capstone project focuses on visualizing geographic differences and quarterly patterns in housing burden and insurance expenditures. The worksheets

complement the descriptive analyses and regression models by presenting the data in visual formats that make comparisons across states and periods easier to explore. The analysis uses HOUSB to measure housing burden and the adjusted insurance expenditure variables AdjVEHINSPQ, AdjHLTHINPQ, AdjLIFINSPQ, and AdjPERINSPQ to represent vehicle, health, life, and personal insurance expenditures. The state-level visualizations focus on geographic variation, while the quarterly visualizations examine how the measures vary across the periods represented by QSQ118.

The state-level map is designed to display average housing burden by state. This visualization provides a geographic perspective on differences in the proportion of income allocated to housing. The map's tooltip includes the state name, average housing burden, and observation count, giving viewers additional information about each state's displayed result. The observation count is important because averages should be interpreted in the context of the number of records contributing to them. States with different sample sizes may have averages estimated from different amounts of data.

The map's interpretation depends on its filters and the accuracy of its state labels. A filter that restricts average housing burden to a selected range can exclude states whose averages fall outside that range. If the goal is to compare the full geographic distribution, the filter should be removed or its purpose clearly explained. States that appear without a displayed value should not automatically be interpreted as having zero housing burden; their data availability and filter status should be checked. In addition, the calculated field used to translate numeric state codes into state names must be verified against the coding scheme in the original Consumer Expenditure Survey data. Until the mapping is verified, the state labels should not be used to make definitive claims about which states have the highest or lowest housing burden.

The state comparison bar chart provides a complementary view of average housing burden across states. Whereas the map emphasizes geographic location, the bar chart makes differences between state averages easier to compare and rank. Presenting the values as bars can help viewers identify the relative size and ordering of the displayed averages. The chart is most informative when state names are accurate, the axis is clearly labeled, and all relevant states can be examined using a consistent scale. If the full set of state comparisons cannot fit legibly in one screenshot, the view should be adjusted or presented in multiple screenshots with the same axis scale. As with the map, conclusions about state rankings should be made only after the state-code mapping has been verified.

The quarterly housing-burden visualization examines average HOUSB across the categories represented by QSQ118. This adds a time-oriented perspective to the state-level analysis by allowing users to examine differences in average housing burden across survey periods. Correct interpretation requires verifying what each QSQ118 value represents and establishing its chronological order. Numeric codes should not automatically be treated as

calendar dates, and a readable quarter label should be used only when its correspondence to the original survey period has been confirmed. Including the observation count in the tooltip provides context for each quarterly average and helps users determine whether the number of contributing records varies across periods.

The quarterly insurance visualization is designed to compare average adjusted vehicle, health, life, and personal insurance expenditures across the same periods. Displaying these outcomes together makes it possible to examine whether their average values follow similar or different patterns across quarters. Because the measures represent different insurance categories, their average expenditure levels may vary substantially. Clearly labeled series and an appropriate vertical-axis title help distinguish the categories and communicate what the chart measures. The results describe average adjusted expenditure per observation, not total spending across all households. Quarterly labels and ordering should be verified before drawing conclusions about changes over time, and the number of observations contributing to each average should be considered when interpreting the trends.

Overall, Tableau contributes geographic and time-oriented visualizations to the capstone project. The state map and bar chart provide complementary approaches to comparing average housing burden across states, while the quarterly charts are intended to show how housing burden and insurance expenditures vary across survey periods. These visualizations differ from Excel's pivot-table summaries, SQL's grouped queries, Python's data-validation and analytical work, and R's regression models. Tableau's contribution is to make geographic and temporal comparisons accessible through visual presentation. The reliability of the resulting interpretations depends on verifying state-code labels, checking filters and missing values, and ensuring that QSQ118 is mapped to the correct chronological periods. Once these details are confirmed, the visualizations can help communicate the project's findings without implying that descriptive differences establish causal relationships.